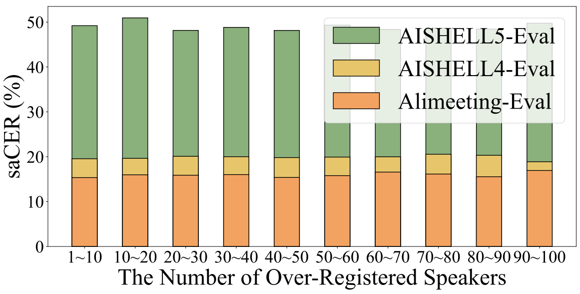

Figure 5: The saCER of SpeakerLM under Over-Regist condition with different numbers of over-registered speakers.

Impact of Multi-Stage Training. Figure 4 illustrates the performance of SpeakerLM across the proposed four different training stages (Section 3.3). On both Alimeeting-Eval and Simulation-Test, the performance of SpeakerLM improves progressively, demonstrating the effectiveness of the proposed multi-stage training strategy. However, on AISHELL4-Eval and AISHELL5-Eval, CERs at Stage 2 are higher than those at Stage 1. This is because Stage 2 relies on simulated data, while the simulation process does not incorporate any audio data from these two datasets, leading to a domain mismatch. With four training stages, SpeakerLM demonstrates strong generalizability on the out-of-domain test set, i.e, AISHELL5-Eval. This suggests that the subsequent training stages (Stage 3 and Stage 4), which involve fine-tuning on more realistic and diverse meeting-style data, are crucial for mitigating domain mismatch and enhancing the model's robustness across different evaluation scenarios. In addition, the ASR performance on simulated data is consistently better than that on real data, which is attributed to the simplicity of the simulated data. Nevertheless, we believe the simulation process remains meaningful, as it closely resembles office or workplace scenarios where a voice assistant is used to record meetings.

5.4 Performance with Speaker Registration

In this section, we explore the SDR performance of SpeakerLM with different numbers of registered speakers.

Match-Regist vs Over-Regist. In Table 3, we first present the performance of SpeakerLM and SA-Transformer on AliMeeting-Eval. Results show that SpeakerLM outperforms SA-Transformer, achieving a 25.98% absolute improvement in saCER. Then, we compare the performance of SpeakerLM under different conditions. For Over-Regist, N_{ov} randomly ranges from 1 to 50, consistent with the training setup. In terms of CER and cpCER, SpeakerLM shows comparable performance across the two registration mechanisms. However, for Δsa, the metrics are consistently lower under the Match-Regist setting, suggesting that the presence of redundant registered speakers introduces unnecessary information, negatively affecting the SD performance.

Impact of the Number of Over-Registered Speakers. Figure 5 explores the impact of the number of over-registered speakers (i.e., N_{ov}) on model performance. We do not observe significant performance degradation as N_{ov} increases. This reflects SpeakerLM's robustness to redundant speaker identities and the ability to focus on relevant speaker representations during inference.

Dataset	System (RM)	CER	saCER	Δsa
Alimeeting-Eval	SA-Transformer (MR)*	-	41.55	-
	SpeakerLM (MR)	13.98	**15.57**	**1.59**
	SpeakerLM (OR)	**13.96**	15.71	1.75
AISHELL4-Eval	SpeakerLM (MR)	**17.13**	**19.73**	**2.60**
	SpeakerLM (OR)	17.15	20.16	3.01
AISHELL5-Eval	SpeakerLM (MR)	47.05	47.36	**0.31**
	SpeakerLM (OR)	**46.69**	**47.35**	0.66

* Due to the lack of an open-source implementation, we adapt the reported results from the original work for comparison.

Table 3: The performance of different systems on AliMeeting-Eval, AISHELL4-Eval and AISHELL5-Eval. "RM", "MR" and "OV" represent "Register Mode", "Match-Regist" and "Over-Regist", respectively.

5.5 Impact of Embedding Extractors

In the experiments above, we adopt ERes2NetV2 as the speaker embedding extractor. To investigate the impact of the speaker embeddings, we replace ERes2NetV2 with another speaker embedding model, i.e., CAM++ (Wang et al. 2023), for training and test. It should be noted that ERes2NetV2 performs better than CAM++ in various speaker verification benchmarks (Chen et al. 2025b). Results in Table 4 show that SpeakerLM using embeddings from ERes2NetV2 outperforms the variant with CAM++, indicating that improvements in the speaker embedding model can further enhance the overall performance of SpeakerLM.

Register Mode	Extractor	CER	cpCER	saCER
No-Regist	CAM++	14.63	16.74	-
	ERes2NetV2	**13.97**	**16.05**	-
Match-Regist	CAM++	14.74	-	17.23
	ERes2NetV2	**13.98**	-	**15.57**
Over-Regist	CAM++	14.71	-	16.92
	ERes2NetV2	**13.96**	-	**15.71**

Table 4: The performance of SpeakerLM on AliMeeting-Eval with different embedding extractors.

6 Conclusions

In this work, we present SpeakerLM, the first MLLM designed for end-to-end SDR. We introduce a flexible registration mechanism into SpeakerLM, enabling the model to perform SDR with different numbers of registered speakers. Compared with strong cascaded baselines built from SOTA SD and ASR models, SpeakerLM achieves superior performance on both in-domain and out-of-domain test sets. This work demonstrates the potential of MLLMs to model SDR in a scalable, data-driven, and autoregressive manner.

References

An, K.; Chen, Q.; Deng, C.; Du, Z.; Gao, C.; Gao, Z.; Gu, Y.; He, T.; Hu, H.; Hu, K.; et al. 2024. FunAudioLLM: Voice Understanding and Generation Foundation Models for Natural Interaction Between Humans and LLMs. *CoRR*.

Anguera, X.; Bozonnet, S.; Evans, N.; Fredouille, C.; Friedland, G.; and Vinyals, O. 2012. Speaker diarization: A review of recent research. *IEEE Transactions on audio, speech, and language processing*, 20(2): 356–370.

Boeddeker, C.; Subramanian, A. S.; Wichern, G.; Haeb-Umbach, R.; and Le Roux, J. 2024. TS-SEP: Joint diarization and separation conditioned on estimated speaker embeddings. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 32: 1185–1197.

Bredin, H. 2023. pyannote. audio 2.1 speaker diarization pipeline: principle, benchmark, and recipe. In *Proc. Interspeech 2023*, 1983–1987. ISCA.

Bredin, H.; Yin, R.; Coria, J. M.; Gelly, G.; Korshunov, P.; Lavechin, M.; Fustes, D.; Titeux, H.; Bouaziz, W.; and Gill, M.-P. 2020. Pyannote. audio: neural building blocks for speaker diarization. In *IEEE International conference on acoustics, speech and signal processing (ICASSP)*, 7124–7128. IEEE.

Chang, Y.; Wang, X.; Wang, J.; Wu, Y.; Yang, L.; Zhu, K.; Chen, H.; Yi, X.; Wang, C.; Wang, Y.; et al. 2024. A survey on evaluation of large language models. *ACM transactions on intelligent systems and technology*, 15(3): 1–45.

Chen, Q.; Chen, Y.; Chen, Y.; Chen, M.; Chen, Y.; Deng, C.; Du, Z.; Gao, R.; Gao, C.; Gao, Z.; et al. 2025a. Minmo: A multimodal large language model for seamless voice interaction. *arXiv preprint arXiv:2501.06282*.

Chen, S.; Wang, C.; Chen, Z.; Wu, Y.; Liu, S.; Chen, Z.; Li, J.; Kanda, N.; Yoshioka, T.; Xiao, X.; et al. 2022. Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6): 1505–1518.

Chen, Y.; Zheng, S.; Wang, H.; Cheng, L.; Chen, Q.; Zhang, S.; and Li, J. 2024. ERes2NetV2: Boosting Short-Duration Speaker Verification Performance with Computational Efficiency. In *Proc. Interspeech 2024*, 3245–3249.

Chen, Y.; Zheng, S.; Wang, H.; Cheng, L.; Zhu, T.; Huang, R.; Deng, C.; Chen, Q.; Zhang, S.; Wang, W.; et al. 2025b. 3D-Speaker-Toolkit: An Open-Source Toolkit for Multimodal Speaker Verification and Diarization. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1–5. IEEE.

Chu, Y.; Xu, J.; Yang, Q.; Wei, H.; Wei, X.; Guo, Z.; Leng, Y.; Lv, Y.; He, J.; Lin, J.; et al. 2024. Qwen2-audio technical report. *arXiv preprint arXiv:2407.10759*.

Cornell, S.; Jung, J.-w.; Watanabe, S.; and Squartini, S. 2024. One model to rule them all? towards end-to-end joint speaker diarization and speech recognition. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 11856–11860. IEEE.

Cornell, S.; Wiesner, M. S.; Watanabe, S.; Raj, D.; Chang, X.; Garcia, P.; Masuyam, Y.; Wang, Z.-Q.; Squartini, S.; and Khudanpur, S. 2023. The CHiME-7 DASR Challenge: Distant Meeting Transcription with Multiple Devices in Diverse Scenarios. In *Proc. CHiME 2023*, 1–6.

Dai, Y.; Wang, H.; Li, X.; Zhang, Z.; Wang, S.; Xie, L.; Xu, X.; Guo, H.; Zhang, S.; Bu, H.; et al. 2025. AISHELL-5: The First Open-Source In-Car Multi-Channel Multi-Speaker Speech Dataset for Automatic Speech Diarization and Recognition. *arXiv preprint arXiv:2505.23036*.

Ding, D.; Ju, Z.; Leng, Y.; Liu, S.; Liu, T.; Shang, Z.; Shen, K.; Song, W.; Tan, X.; Tang, H.; et al. 2025. Kimi-audio technical report. *arXiv preprint arXiv:2504.18425*.

Du, J.; Na, X.; Liu, X.; and Bu, H. 2018. Aishell-2: Transforming mandarin asr research into industrial scale. *arXiv preprint arXiv:1808.10583*.

Efstathiadis, G.; Yadav, V.; and Abbas, A. 2025. LLM-based speaker diarization correction: A generalizable approach. *Speech Communication*, 170: 103224.

Fu, Y.; Cheng, L.; Lv, S.; Jv, Y.; Kong, Y.; Chen, Z.; Hu, Y.; Xie, L.; Wu, J.; Bu, H.; et al. 2021. AISHELL-4: An Open Source Dataset for Speech Enhancement, Separation, Recognition and Speaker Diarization in Conference Scenario. In *Proc. Interspeech*, 3665–3669.

Fujita, Y.; Kanda, N.; Horiguchi, S.; Xue, Y.; Nagamatsu, K.; and Watanabe, S. 2019. End-to-end neural speaker diarization with self-attention. In *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 296–303. IEEE.

Gao, M.; Wu, S.; Chen, H.; Du, J.; Lee, C.-H.; Watanabe, S.; Chen, J.; Marco, S. S.; and Scharenborg, O. 2025. The multimodal information based speech processing (misp) 2025 challenge: Audio-visual diarization and recognition. *arXiv preprint arXiv:2505.13971*.

Gao, Z.; Zhang, S.; McLoughlin, I.; and Yan, Z. 2022. Paraformer: Fast and Accurate Parallel Transformer for Non-autoregressive End-to-End Speech Recognition. In *Proc. Interspeech*, 2063–2067.

Han, J.; Landini, F.; Rohdin, J.; Silnova, A.; Diez, M.; and Burget, L. 2025a. Leveraging self-supervised learning for speaker diarization. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1–5. IEEE.

Han, J.; Landini, F.; Rohdin, J.; Silnova, A.; Diez, M.; Cernocky, J.; and Burget, L. 2025b. Fine-tune Before Structured Pruning: Towards Compact and Accurate Self-Supervised Models for Speaker Diarization. *Proc. Interspeech 2025*.

Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; Chen, W.; et al. 2022. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2): 3.

Kinoshita, K.; Delcroix, M.; and Tawara, N. 2021a. Advances in Integration of End-to-End Neural and Clustering-Based Diarization for Real Conversational Speech. In *Proc. Interspeech 2021*, 3565–3569.

Kinoshita, K.; Delcroix, M.; and Tawara, N. 2021b. Integrating end-to-end neural and clustering-based diarization: Getting the best of both worlds. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 7198–7202. IEEE.

Landini, F.; Lozano-Diez, A.; Diez, M.; and Burget, L. 2022. From Simulated Mixtures to Simulated Conversations as Training Data for End-to-End Neural Diarization. In *Proc. Interspeech*, 5095–5099.

Liang, Y.; Shi, M.; Yu, F.; Li, Y.; Zhang, S.; Du, Z.; Chen, Q.; Xie, L.; Qian, Y.; Wu, J.; et al. 2023. The Second Multi-Channel Multi-Party Meeting Transcription Challenge (M2MeT 2.0): A Benchmark for Speaker-Attributed ASR. In *IEEE Automatic Speech Recognition and Understanding Workshop, ASRU*, 10389625. IEEE.

Meng, L.; Hu, S.; Kang, J.; Li, Z.; Wang, Y.; Wu, W.; Wu, X.; Liu, X.; and Meng, H. 2025. Large language model can transcribe speech in multi-talker scenarios with versatile instructions. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1–5. IEEE.

O'Shaughnessy, D. 2025. Speaker Diarization: A Review of Objectives and Methods. *Applied Sciences*, 15(4): 2002.

Panayotov, V.; Chen, G.; Povey, D.; and Khudanpur, S. 2015. Librispeech: an asr corpus based on public domain audio books. In *International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 5206–5210. IEEE.

Park, T. J.; Dhawan, K.; Koluguri, N.; and Balam, J. 2024. Enhancing speaker diarization with large language models: A contextual beam search approach. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 10861–10865. IEEE.

Park, T. J.; Kanda, N.; Dimitriadis, D.; Han, K. J.; Watanabe, S.; and Narayanan, S. 2022. A review of speaker diarization: Recent advances with deep learning. *Computer Speech & Language*, 72: 101317.

Raj, D.; Denisov, P.; Chen, Z.; Erdogan, H.; Huang, Z.; He, M.; Watanabe, S.; Du, J.; Yoshioka, T.; Luo, Y.; et al. 2021. Integration of speech separation, diarization, and recognition for multi-speaker meetings: System description, comparison, and analysis. In *IEEE spoken language technology workshop (SLT)*, 897–904. IEEE.

Singh, A. 2025. Consequences of the Turing Test: OpenAI's GPT-4.5. *Available at SSRN 5205937*.

Team, Q. 2024. Qwen2.5: A Party of Foundation Models.

Tranter, S. E.; and Reynolds, D. A. 2006. An overview of automatic speaker diarization systems. *IEEE Transactions on audio, speech, and language processing*, 14(5): 1557–1565.

Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, Ł.; and Polosukhin, I. 2017. Attention is all you need. *Advances in neural information processing systems*, 30.

Wang, H.; Zheng, S.; Chen, Y.; Cheng, L.; and Chen, Q. 2023. CAM++: A Fast and Efficient Network for Speaker Verification Using Context-Aware Masking. In *Proc. Interspeech 2023*, 5301–5305.

Wang, Q.; Huang, Y.; Zhao, G.; Clark, E.; Xia, W.; and Liao, H. 2024. DiarizationLM: Speaker Diarization Post-Processing with Large Language Models. *CoRR*.

Wei, J.; Tay, Y.; Bommasani, R.; Raffel, C.; Zoph, B.; Borgeaud, S.; Yogatama, D.; Bosma, M.; Zhou, D.; Metzler, D.; et al. 2022. Emergent Abilities of Large Language Models. *Transactions on Machine Learning Research*.

Yu, F.; Zhang, S.; Guo, P.; Fu, Y.; Du, Z.; Zheng, S.; Huang, W.; Xie, L.; Tan, Z.-H.; Wang, D.; et al. 2022. Summary on the ICASSP 2022 multi-channel multi-party meeting transcription grand challenge. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 9156–9160. IEEE.

Zhou, P.; Xie, X.; Lin, Z.; and Yan, S. 2024. Towards understanding convergence and generalization of AdamW. *IEEE transactions on pattern analysis and machine intelligence*.